06 — Metadata forensics

What it detects: signs a PDF was edited, re-exported or assembled after it was created: editing software fingerprints, inconsistent or missing timestamps, incremental-save revisions, XMP edit history that contradicts the file, orphaned objects, embedded JavaScript, and a stripped metadata block.

Code: `services/forensics/metadata_forensics.py` (`analyze_pdf_metadata`), run by `tasks/metadata_forensics_task.py` (`run_metadata_forensics`); stores `check_type = metadata_forensics` .

What is read

Using **pikepdf** on the PDF's own object model (never a page render):

- the **Info dictionary** (`/Title /Author /Creator /Producer /CreationDate /ModDate`);
- the separate **XMP metadata stream**, including custom namespaces and the `xmpMM:History` revision chain;
- the **document ID chain** (`/ID`);
- the file's real **incremental-update / xref-revision structure** (`_walk_xref_chain`);
- **reachability**: which objects are unreachable from `/Root` / `/Info` (orphans);
- JavaScript / `OpenAction` , optional content groups, and digital-signature fields.

Findings

Every finding is `{finding, severity, description, data?}` . `severity` is `info | low | medium | high` . `info` findings are context (they show the raw metadata) and never affect the result.

`finding`	Severity	Meaning
`pdf_unreadable`	high	The file could not be opened as a PDF
`metadata_entirely_stripped`	high	No Info and no XMP metadata at all
`editing_software_detected`	high	Producer/Creator/ `softwareAgent` matches an image/graphics editor (`METADATA_FORENSICS_EDITING_SOFTWARE_NAMES` : photoshop, gimp, illustrator, affinity, coreldraw, paint.net, pixlr, canva, inkscape)
`pdf_editor_detected`	medium	Producer/Creator names an online or desktop PDF editor (`METADATA_FORENSICS_PDF_EDITOR_NAMES` : iLovePDF, Smallpdf, Sejda, PDFescape, PDF-XChange Editor, PhantomPDF / Foxit PDF Editor, Nitro Pro, PDFelement, Soda PDF, pdfFiller, DocHub, PDF Candy). One finding per tool. Print drivers of the same vendors are not listed
`mod_date_after_creation_date`	high	ModDate later than CreationDate by more than `METADATA_FORENSICS_MOD_DATE_THRESHOLD_SECONDS` (default 1 s — only jitter tolerance; being modified at all is the anomaly)
`info_xmp_creation_date_mismatch`	high	Info and XMP creation dates differ by > 1 minute

finding	Severity	Meaning
info_xmp_mod_date_mismatch	high	Info and XMP modification dates differ by > 1 minute
info_xmp_producer_mismatch	medium	Info /Producer ≠ XMP producer
incremental_updates_present	medium (1 update) / high (> 1)	The file has been saved incrementally
history_editing_tool_anomaly	medium	An XMP history entry's softwareAgent names an image editor or a PDF editor
history_scanned_document_edited	high	An XMP history entry's action is editedScannedDoc — Acrobat's "Edit scanned document": the scan was converted to editable text and edited
history_edit_after_final_date	high	An XMP history event is later than the file's modified date (+1 min)
history_shorter_than_revisions	medium	Fewer history entries than xref revisions
history_absent_despite_revisions	medium	Revisions exist but the history is empty
orphaned_objects	medium (orphans found) / info	Content objects present but unreachable from /Root or /Info — typical of deleted content. Only content counts: streams, fonts, images and form XObjects, pages, annotations (data.kinds). A bare array, number or plain dictionary (macOS/iOS Quartz leaves its page box [0 0 612 792]), or an empty stream (/Length ≤ 16 or nothing once decoded — PDFium leaves an empty Form XObject per page) is not counted (data.trivial_ignored)
javascript_or_openaction	medium (flagged) / info	Embedded JS or auto-run action
optional_content_groups	low / info	Layers present

`finding`	Severity	Meaning
`editable_text_over_scan`	medium	Page structure (`services/forensics/page_structure.py` , from the drawing order): an image covering ≥ 90 % of the page drawn **before** visible live text (≥ 15 runs spanning ≥ 50 % of the page height) — a scan converted to editable text, **whatever the fonts are called** (a converter's `-NNNN` subsets, listed in `data.converter_font_pages` , or ordinary `ABCDEF+` subsets as Acrobat's editor leaves them). Does not depend on XMP, so it still holds when the metadata is stripped. An image drawn *over* the text (a flattened page pasted on top) is not this structure. A normal searchable scan keeps its OCR text invisible (render mode 3). Such pages also get the ghost-content check (07a)
`producer_scrubbed`	medium	The XMP packet was written by Adobe's own library (`x:xmptk` "Adobe XMP Core …", read from the raw stream), yet Info Producer/Creator, XMP Producer/CreatorTool and the edit history are all empty. Adobe software always records its name and saves there, so they were removed afterwards. A fully empty file is `metadata_entirely_stripped` instead
`web_page_origin`	low / medium	The producer is a browser's or an HTML engine's PDF output (PDFium, Skia/PDF, Chrome, wkhtmltopdf, Puppeteer, Playwright, WeasyPrint, Prince…): the document was built as a web page and printed. **Medium** (`metadata.web_page_origin` , 10) when it also has no CreationDate or uses ≥ 3 stock colours of a CSS framework (Bootstrap 5, Tailwind — `services/forensics/pdf_facts.py`); **low** (shown only) with the engine alone, since school portals print from browsers too
`digital_signature`	info	Signature field present / signed / unsigned / absent
`info_dictionary` , `xmp_metadata` , `xmp_history` , `revision_count` , `document_id_chain`	info	Raw context

The check `result` is `flag` **if any finding is** `medium` **or** `high` , else `pass` .

All `metadata.*` rules together add at most the company's metadata cap (default 40) to a case's score — see 11.

Output shape (real, truncated)

```
{
  "result": "flag",
  "details": [
    {"finding": "orphaned_objects", "severity": "medium",
     "description": "Found 25 object(s) present in the file but not reachable from /Root or /Info — likely
left…",
     "data": {"count": 25, "object_numbers": [2, 3, "…"]}},
    {"finding": "info_dictionary", "severity": "info", "description": "Info dictionary has 6 field(s).",
     "data": {"/Creator": "Adobe Photoshop 2024 (Windows)", "/ModDate": "D:20260902164500Z",
              "/CreationDate": "D:20260801090000Z", "/Producer": "Adobe Photoshop 2024 (Windows)", "…":
"…"}}
  ]
}
```

`details` is a **list** of findings (the same shape ELA, copy-move, duplicate and visual review use, so one findings-list UI renders them all). These findings carry no `page` / `bounding_box` ; metadata is not positional.

Known limitations

- **PDF only.** Images and Word files are not analyzed.
- Many legitimate tools leave incremental updates or editor fingerprints (a scanned-then-annotated invoice, for example). A finding is a reason to look, not proof.
- A determined forger can rewrite or strip metadata; `metadata_entirely_stripped` is itself scored because clean-room generators rarely strip everything.
- Uploads are PDF-only, so every new document is analysed; a non-PDF stored before that rule is a quiet no-op.
- The visual-review task cross-references this result (`metadata_forensics_correlation`) to strengthen or explain its own findings; that cross-reference is deliberately **not scored** to avoid double counting.

Risk rules fed

`rule_id`	Finding	Fires on	Weight	Severity
`metadata.pdf_unreadable`	`pdf_unreadable`	any	20	high
`metadata.stripped`	`metadata_entirely_stripped`	any	12	medium
`metadata.editing_software_detected`	`editing_software_detected`	any	30	high
`metadata.creation_date_mismatch`	`info_xmp_creation_date_mismatch`	any	18	high
`metadata.mod_date_mismatch`	`info_xmp_mod_date_mismatch`	any	18	high
`metadata.producer_mismatch`	`info_xmp_producer_mismatch`	any	10	medium
`metadata.pdf_editor_producer`	`pdf_editor_detected`	any	10	medium
`metadata.modified_after_creation`	`mod_date_after_creation_date`	any	15	high
`metadata.incremental_update_single`	`incremental_updates_present`	medium	8	medium
`metadata.incremental_update_multiple`	`incremental_updates_present`	high	20	high
`metadata.history_tool_anomaly`	`history_editing_tool_anomaly`	any	12	medium
`metadata.history_scanned_doc_edited`	`history_scanned_document_edited`	any	15	high
`metadata.editable_text_over_scan` (v2: any font naming)	`editable_text_over_scan`	medium, high	10	medium
`metadata.producer_scrubbed`	`producer_scrubbed`	any	10	medium

rule_id	Finding	Fires on	Weight	Severity
`metadata.web_page_origin`	`web_page_origin`	medium, high	10	medium
`metadata.history_edit_after_final`	`history_edit_after_final_date`	any	22	high
`metadata.history_shorter_than_revisions`	`history_shorter_than_revisions`	any	8	medium
`metadata.history_absent_despite_revisions`	`history_absent_despite_revisions`	any	8	medium
`metadata.orphaned_objects` (v3: content only, empty streams ignored)	`orphaned_objects`	medium, high	8	medium
`metadata.javascript_or_openaction`	`javascript_or_openaction`	medium, high	10	medium

Note the two severities: a rule's `severity` (how it is displayed) is independent of the *finding's* severity (which the rule may filter on with `severity_in`).